

ALGORITHM LongestBalancedSubstring(s)

INPUT: A string s of lowercase letters

OUTPUT: Length of the longest substring containing exactly 2

BEGIN

n = LENGTH(s)

maxLen = 0

FOR i = 0 TO n - 1

 count[0..25] = {0}

 FOR j = i TO n - 1 DO

 count[s[j] - 'a'] = count[s[j] - 'a'] + 1

 distinct = 0

 first = -1

 second = -1

 FOR k = 0 TO 25

 IF count[k] > 0 THEN

 distinct = distinct + 1

 IF first = -1 THEN

 first = count[k]

 ELSE

 second = count[k]

 END IF

 END IF

 END FOR

 IF distinct = 2 AND first = second THEN

 len = j - i + 1

 IF len > maxLen THEN

 maxLen = len

 END IF

```
        END IF
    END FOR
END FOR
RETURN maxLen
END
```